

HYDRANAL™-Coulomat AG-H**34843-500ML-US**

Version 1.2

Revision Date 08/20/2018

Print Date 10/17/2019

SECTION 1. IDENTIFICATION

Product name : HYDRANAL™-Coulomat AG-H

Number : 000000020626

Product Use Description : Laboratory chemicals
Scientific research and development

Manufacturer or supplier's details : Honeywell International Inc.
1953 South Harvey Street
Muskegon, MI 49442

For more information call : 1-800-368-0050
+1-231-726-3171

(Monday-Friday, 9:00am-5:00pm)

In case of emergency call : **Medical: 1-800-498-5701 or +1-303-389-1414**
: **Transportation (CHEMTREC): 1-800-424-9300 or**
: **+1-703-527-3887**
:
: (24 hours/day, 7 days/week)

SECTION 2. HAZARDS IDENTIFICATION**Emergency Overview**

Form : liquid

Color : light yellow

Odor : aromatic

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Classification of the substance or mixture

Classification of the substance or mixture : Flammable liquids, Category 2
Acute toxicity, Category 4, Inhalation
Skin corrosion, Category 1B
Serious eye damage, Category 1
Reproductive toxicity, Category 1B
Specific target organ toxicity - single exposure, Category 1, Eyes, Nervous system, Systemic toxicity
Specific target organ toxicity - single exposure, Category 3, Respiratory system

GHS Label elements, including precautionary statements

Symbol(s)



Signal word

: Danger

Hazard statements

: Highly flammable liquid and vapour.
Causes severe skin burns and eye damage.
Harmful if inhaled.
May cause respiratory irritation.
May damage fertility or the unborn child.
Causes damage to organs.

Precautionary statements

: **Prevention:**
Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
Keep away from heat/sparks/open flames/hot surfaces. No smoking.
Keep container tightly closed.
Ground/bond container and receiving equipment.
Use explosion-proof electrical/ ventilating/ lighting/ equipment.

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Use only non-sparking tools.
Take precautionary measures against static discharge.
Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
Wash skin thoroughly after handling.
Do not eat, drink or smoke when using this product.
Use only outdoors or in a well-ventilated area.
Wear protective gloves/protective clothing/eye protection/face protection.

Response:

IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower.
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Immediately call a POISON CENTER/doctor.
Wash contaminated clothing before reuse.
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

Storage:

Store in a well-ventilated place. Keep container tightly closed.
Keep cool.
Store locked up.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

Carcinogenicity

No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP, IARC, or OSHA.

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SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical nature : Mixture

Chemical name	CAS-No.	Concentration
Methanol	67-56-1	>=40.00 - <60.00 %
N,N-Dimethyldodecan-1-amine	112-18-5	>=10.00 - <20.00 %
1-Pentanol	71-41-0	>=15.00 - <25.00 %
Imidazole	288-32-4	>=5.00 - <10.00 %
Imidazol hydrobromide	101023-55-6	>=5.00 - <10.00 %
Sulphur dioxide	7446-09-5	>=5.00 - <10.00 %
1H-Imidazole monohydriodide	68007-08-9	<5.00 %

SECTION 4. FIRST AID MEASURES

General advice : First aider needs to protect himself. Move out of dangerous area. Take off all contaminated clothing immediately.

Inhalation : Remove to fresh air. Keep patient warm and at rest. Call a physician immediately.

Skin contact : Wash off immediately with plenty of water. If skin irritation persists, call a physician.

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- Eye contact : In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Protect unharmed eye.
- Ingestion : Call a physician immediately. Do NOT induce vomiting. When swallowed, allow water to be drunk.

SECTION 5. FIREFIGHTING MEASURES

- Suitable extinguishing media : Water spray
Foam
Carbon dioxide (CO₂)
Dry powder
- Unsuitable extinguishing media : Do not use a solid water stream as it may scatter and spread fire.
- Specific hazards during firefighting : Flammable.
Vapours may form explosive mixtures with air.
Vapours are heavier than air and may spread along floors.
Vapors may travel to areas away from work site before igniting/flashing back to vapor source.
In case of fire hazardous decomposition products may be produced such as:
Carbon monoxide
Carbon dioxide (CO₂)
nitrogen oxides (NO_x)
Sulphur oxides
Hydrogen halides
- Special protective equipment for firefighters : Wear an approved positive pressure self-contained breathing apparatus in addition to standard fire fighting gear.
- Further information : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

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SECTION 6. ACCIDENTAL RELEASE MEASURES

- | | | |
|---|---|---|
| Personal precautions, protective equipment and emergency procedures | : | Evacuate personnel to safe areas.
Wear personal protective equipment. Unprotected persons must be kept away.
Remove all sources of ignition.
Ensure adequate ventilation. |
| Environmental precautions | : | Should not be released into the environment. |
| Methods and materials for containment and cleaning up | : | Ventilate the area.
No sparking tools should be used.
Use explosion-proof equipment.
Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13). |

SECTION 7. HANDLING AND STORAGE**Handling**

- | | | |
|---|---|---|
| Precautions for safe handling | : | Wear personal protective equipment.
Use only in well-ventilated areas. |
| Advice on protection against fire and explosion | : | Keep away from sources of ignition - No smoking.
Take measures to prevent the build up of electrostatic charge.
The heavy vapours can overcome a considerable distance up to the source of ignition.
Vapours may form explosive mixtures with air. |

Storage

- | | | |
|--|---|--|
| Conditions for safe storage, including any incompatibilities | : | Store in area designed for storage of flammable liquids. Protect from physical damage.
Keep containers tightly closed in a dry, cool and well-ventilated place. |
|--|---|--|

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Store in original container.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

- Protective measures : Ensure that eyewash stations and safety showers are close to the workstation location.
Legal requirements are to be considered in regard of the selection, use and care of personal protective equipment.
Do not breathe vapours or spray mist.
- Engineering measures : Use with local exhaust ventilation.
Prevent vapour buildup by providing adequate ventilation during and after use.
- Eye protection : Safety goggles
- Hand protection : Impervious butyl rubber gloves
Gloves must be inspected prior to use.
Replace when worn.
- Skin and body protection : Protective suit
- Respiratory protection : In case of insufficient ventilation wear suitable respiratory equipment.
Recommended Filter type:
AX: Organic gas and low boiling vapour type
- Hygiene measures : Take off all contaminated clothing immediately.
Remove and wash contaminated clothing before re-use.
Wash hands before breaks and at the end of workday.
When using do not eat or drink.

Exposure Guidelines

Components	CAS-No.	Value	Control parameters	Update	Basis
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Methanol	67-56-1	SKIN_DE S : Skin designati on:	Can be absorbed through the skin.	2008	ACGIH:US. ACGIH Threshold Limit Values
Methanol	67-56-1	STEL : Short term exposure limit	(250 ppm)	2008	ACGIH:US. ACGIH Threshold Limit Values
Methanol	67-56-1	TWA : Time weighted average	(200 ppm)	2008	ACGIH:US. ACGIH Threshold Limit Values
Methanol	67-56-1	STEL : Short term exposure limit	325 mg/m3 (250 ppm)	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards
Methanol	67-56-1	SKIN_DE S : Skin designati on:	Can be absorbed through the skin.	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards
Methanol	67-56-1	REL : Recomm ended exposure limit (REL):	260 mg/m3 (200 ppm)	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards

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Methanol	67-56-1	PEL : Permissible exposure limit	260 mg/m3 (200 ppm)	02 2006	OSHA_TRANS:US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)
Methanol	67-56-1	SKIN_FINAL : Skin designation (Final Rule Limit applies):	Can be absorbed through the skin.	1989	Z1A:US. OSHA Table Z-1-A (29 CFR 1910.1000)
Methanol	67-56-1	STEL : Short term exposure limit	325 mg/m3 (250 ppm)	1989	Z1A:US. OSHA Table Z-1-A (29 CFR 1910.1000)
Methanol	67-56-1	TWA : Time weighted average	260 mg/m3 (200 ppm)	1989	Z1A:US. OSHA Table Z-1-A (29 CFR 1910.1000)
1-Pentanol	71-41-0	TWA : Time weighted average	360 mg/m3 (100 ppm)	2007	WEEL:US. OARS. WEELs Workplace Environmental Exposure Level Guide
Sulphur dioxide	7446-09-5	STEL : Short term exposure limit	(0.25 ppm)	2009	ACGIH:US. ACGIH Threshold Limit Values

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Sulphur dioxide	7446-09-5	STEL : Short term exposure limit	13 mg/m3 (5 ppm)	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards
Sulphur dioxide	7446-09-5	REL : Recomm ended exposure limit (REL):	5 mg/m3 (2 ppm)	2005	NIOSH/GUIDE:US. NIOSH: Pocket Guide to Chemical Hazards
Sulphur dioxide	7446-09-5	PEL : Permissi ble exposure limit	13 mg/m3 (5 ppm)	02 2006	OSHA_TRANS:US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)
Sulphur dioxide	7446-09-5	STEL : Short term exposure limit	13 mg/m3 (5 ppm)	1989	Z1A:US. OSHA Table Z-1-A (29 CFR 1910.1000)
Sulphur dioxide	7446-09-5	TWA : Time weighted average	5 mg/m3 (2 ppm)	1989	Z1A:US. OSHA Table Z-1-A (29 CFR 1910.1000)

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Color : light yellow

Odor : aromatic

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Odor threshold	: Note: no data available
pH	: 5.0 - 7.0 at , 20 °C
Melting point/range	: Note: no data available
Boiling point/boiling range	: 63 °C at 1,013 hPa
Flash point	: 57 °F (14 °C)
Evaporation rate	: Note: no data available
Lower explosion limit	: Note: no data available
Upper explosion limit	: Note: no data available
Vapor pressure	: Note: no data available
Vapor density	: Note: no data available
Density	: 0.930 g/cm3 at 20 °C
Water solubility	: Note: completely miscible
Partition coefficient: n-octanol/water	: Note: no data available

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Ignition temperature : Note: no data available

Decomposition temperature : Note: No decomposition if used as directed.

Viscosity, dynamic : Note: no data available

Viscosity, kinematic : Note: no data available

SECTION 10. STABILITY AND REACTIVITY

Chemical stability : Stable under recommended storage conditions.

Possibility of hazardous reactions : Vapours may form explosive mixture with air.
Hazardous polymerisation does not occur.

Conditions to avoid : Heat, flames and sparks.

Incompatible materials : Oxidizing agents

Hazardous decomposition products : In case of fire hazardous decomposition products may be produced such as:
Carbon monoxide
Carbon dioxide (CO₂)
nitrogen oxides (NO_x)
Sulphur oxides
Hydrogen halides

SECTION 11. TOXICOLOGICAL INFORMATION

Acute oral toxicity
Methanol : LD50: 5,628 mg/kg
Species: Rat

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N,N-Dimethyldodecan-1-amine : LD50: 1,220 mg/kg
Species: Rat, male
Method: OECD Test Guideline 401

1-Pentanol : LD50: 3,030 mg/kg
Species: Rat, male and female

Imidazole : LD50: 970 mg/kg
Species: Rat

Imidazol hydrobromide : LD50: > 2,000 mg/kg
Species: Rat
Method: OECD

1H-Imidazole monohydriodide : LD50: > 300 mg/kg
Species: Rat
Method: OECD 423

Acute inhalation toxicity
Methanol : LC50: 64000 ppm
Exposure time: 4 h
Species: Rat

Acute dermal toxicity
Methanol : LD50: 15,800 mg/kg
Species: Rabbit

1-Pentanol : LD50: 2,292 mg/kg
Species: Rabbit, male
Method: OECD Test Guideline 402

1H-Imidazole monohydriodide : LD50: > 2,000 mg/kg
Species: Rat
Method: OECD Test Guideline 402

Skin irritation : Result: Corrosive
Note: The product has not been tested. The information is

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derived from the properties of the individual components.

Eye irritation : Note: no data available

Sensitisation : Note: no data available

Repeated dose toxicity : Note: no data available

Genotoxicity in vitro
Methanol : Note: In vitro tests did not show mutagenic effects

N,N-Dimethyldodecan-1-amine : Test Method: Ames test
Result: negative

1-Pentanol : Test Method: Micronucleus test
Cell type: Chinese hamster fibroblasts
Metabolic activation: without metabolic activation
Result: negative

Imidazole : Test Method: In vitro mammalian cell gene mutation test
Cell type: Chinese hamster fibroblasts
Metabolic activation: with and without metabolic activation
Result: negative
Method: OECD Test Guideline 476

1H-Imidazole
monohydriodide : Result: negative
Method: Mutagenicity (Escherichia coli - reverse mutation
assay)

: Test Method: reverse mutation assay
Cell type: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Result: negative

: Test Method: reverse mutation assay
Cell type: Escherichia coli
Metabolic activation: without metabolic activation

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Result: negative

- : Test Method: comet assay
Cell type: Chinese hamster fibroblasts
Metabolic activation: without metabolic activation
Result: negative
- : Test Method: reverse mutation assay
Cell type: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Result: negative
- : Test Method: Ames test
Result: negative
- : Test Method: Chromosome aberration test in vitro
Cell type: Chinese hamster cells
Result: negative
Method: OECD Test Guideline 473

Genotoxicity in vivo

Methanol

- : Note: In vivo tests did not show mutagenic effects

N,N-Dimethyldodecan-1-amine

- : Test Method: Chromosome aberration test
Species: Mouse, male and female
Cell type: Micronucleus
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Imidazole

- : Test Method: Micronucleus test
Species: Mouse, male and female
Cell type: Bone marrow
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

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Teratogenicity
Imidazole

: Species: Rat Application Route: Oral

No observed adverse effect level: 60 mg/kg body weight

No observed adverse effect level: 60 mg/kg body weight

Method: OECD Test Guideline 414

Result: Embryotoxic effects and adverse effects on the
offspring were detected.**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

Toxicity to fish

: Note: Toxic to aquatic organisms, may cause long-term
adverse effects in the aquatic environment.
The product has not been tested. The information is derived
from the properties of the individual components.Toxicity to daphnia and other
aquatic invertebrates: Note: Toxic to aquatic organisms, may cause long-term
adverse effects in the aquatic environment.
The product has not been tested. The information is derived
from the properties of the individual components.

Toxicity to algae

: Note: Toxic to aquatic organisms, may cause long-term
adverse effects in the aquatic environment.
The product has not been tested. The information is derived
from the properties of the individual components.Toxicity to bacteria
Methanol: EC50: 43,000 mg/l
Exposure time: 5 min
Species: Photobacterium phosphoreumEC50: 40,000 mg/l
Exposure time: 15 min
Species: Photobacterium phosphoreum

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EC50: 39,000 mg/l
Exposure time: 25 min
Species: Photobacterium phosphoreum

1H-Imidazole
monohydriodide : Respiration inhibition
EC50: > 1,000 mg/l
Exposure time: 3 h
Species: activated sludge
Method: OECD 209

Respiration inhibition
NOEC: 320 mg/l
Exposure time: 3 h
Species: activated sludge
Method: OECD 209

Elimination information (persistence and degradability)

Biodegradability : Note: no data available

Further information on ecology

Additional ecological information : Do not flush into surface water or sanitary sewer system.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods : Observe all Federal, State, and Local Environmental regulations.

SECTION 14. TRANSPORT INFORMATION

DOT UN/ID No. : UN 1992
Proper shipping name : FLAMMABLE LIQUID, TOXIC, N.O.S.
(Methanol, Dodecyldimethylamine)

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Class	3
Packing group	II
Hazard Labels	3 (6.1)

IATA	UN/ID No.	: UN 1992
	Description of the goods	: FLAMMABLE LIQUID, TOXIC, N.O.S. (Methanol, Dodecyldimethylamine)
	Class	: 3
	Packaging group	: II
	Hazard Labels	: 3 (6.1)
	Packing instruction (cargo aircraft)	: 364
	Packing instruction (passenger aircraft)	: 352
	Packing instruction (passenger aircraft)	: Y341
IMDG	UN/ID No.	: UN 1992
	Description of the goods	: FLAMMABLE LIQUID, TOXIC, N.O.S. (METHANOL, DODECYLDIMETHYLAMINE)
	Class	: 3
	Packaging group	: II
	Hazard Labels	: 3 (6.1)
	EmS Number	: F-E, S-D
	Marine pollutant	: yes
	IMDG Code segregation group according chapter 3.1.4.4	: NONE,

SECTION 15. REGULATORY INFORMATION**Inventories**

US. Toxic Substances Control Act	: All chemical substances in this product are either listed on the TSCA Inventory or are in compliance with a TSCA Inventory exemption.
China. Inventory of Existing Chemical Substances	: On the inventory, or in compliance with the inventory
Note	: Note: Because of the potential specific inventory listing of

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components of this product line, further, more detailed information can be requested from
SafetyDataSheet@Honeywell.com.

National regulatory information

TSCA : This material must be used in compliance with the TSCA Research and Development Exemption requirements (40 CFR 720.36).

US. EPA CERCLA Hazardous Substances (40 CFR 302) : The following component(s) of this product is/are subject to release reporting under 40 CFR 302 when release exceeds the Reportable Quantity (RQ):

Reportable quantity: 5000 lbs
: Methanol 67-56-1

Reportable quantity: 500 lbs
: Sulphur dioxide 7446-09-5

SARA 302 Components : The following components are subject to reporting levels established by SARA Title III, Section 302:
: Sulphur dioxide 7446-09-5

SARA 313 Components : The following components are subject to reporting levels established by SARA Title III, Section 313:
: Methanol 67-56-1

SARA 311/312 Hazards : Fire Hazard
Acute Health Hazard
Chronic Health Hazard

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California Prop. 65

:



WARNING: This product can expose you to chemicals, listed below, known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Sulphur dioxide	7446-09-5
Methanol	67-56-1

Massachusetts RTK

:

Methanol	67-56-1
1-Pentanol	71-41-0
Sulphur dioxide	7446-09-5

New Jersey RTK

:

Sulphur dioxide	7446-09-5
1-Pentanol	71-41-0
Methanol	67-56-1

Pennsylvania RTK

:

Methanol	67-56-1
1-Pentanol	71-41-0
Sulphur dioxide	7446-09-5

SECTION 16. OTHER INFORMATION

	HMIS III	NFPA
Health hazard	: 3*	3
Flammability	: 3	3
Physical Hazard	: 0	
Instability	:	0

* - Chronic health hazard

Hazard rating and rating systems (e.g. HMIS® III, NFPA): This information is intended solely for the use of individuals trained in the particular system.

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a

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warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. Final determination of suitability of any material is the sole responsibility of the user. This information should not constitute a guarantee for any specific product properties.

Changes since the last version are highlighted in the margin. This version replaces all previous versions.

Previous Issue Date: 08/09/2017

Prepared by Honeywell Performance Materials and Technologies Product Stewardship Group